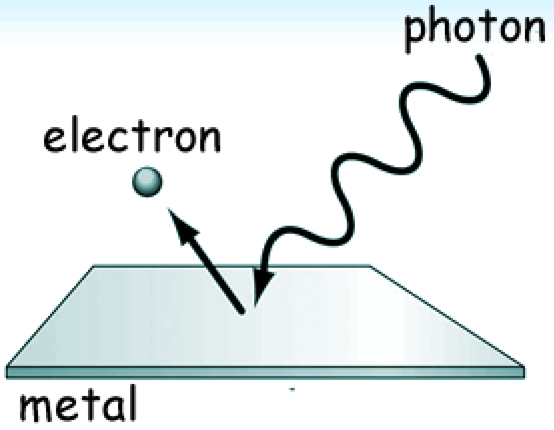


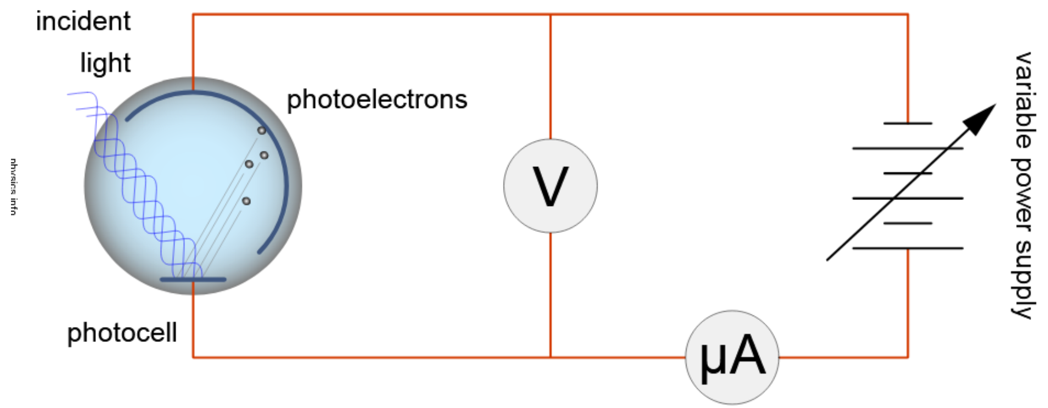
11.2 PHOTOELECTRIC EFFECT

Photoelectric Effect

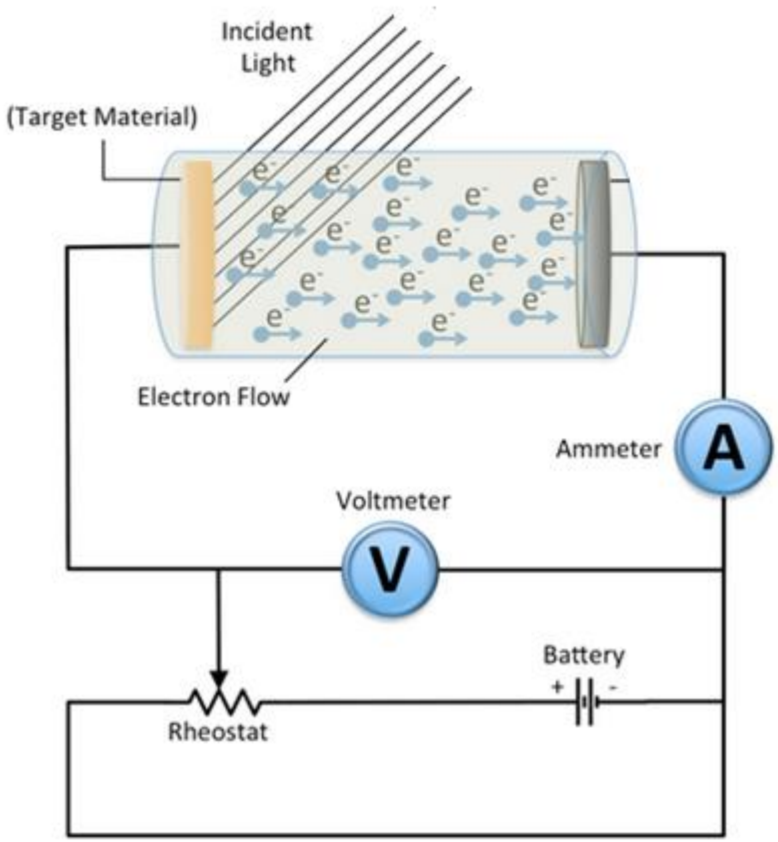
- When an electron in a material absorbs a high enough energy photon, it gains enough kinetic energy to escape from the substance. This is called the photoelectric effect

- The photoelectric effect was first observed in 1887 by Heinrich Hertz during experiments with a spark gap generator (the earliest device that could be called a radio).
- German scientist Philip Leonard investigated further

In 1902 and verified



Apperatus



Incident Light
(Target Material)
Electron Flow
Ammeter
A
Voltmeter
V
Battery
+
-
Rheostat

Observations

- Existence of threshold frequency- No photoelectrons are emitted if the frequency is lower than a certain frequency. No matter the intensity of the incident radiation is or for how long it falls on the surface of the metal.
- Emission is instantaneous - Emission of photoelectrons takes place almost instantaneously after the light shines on the metal, with no detectable time delay. It does not depend on the intensity of the incident radiation.
- Maximum Kinetic Energy of the photoelectron is independent of intensity - The maximum kinetic is dependent linearly on the frequency of the radiation and is independent of its intensity.
- Rate of emission of photoelectrons proportional to intensity

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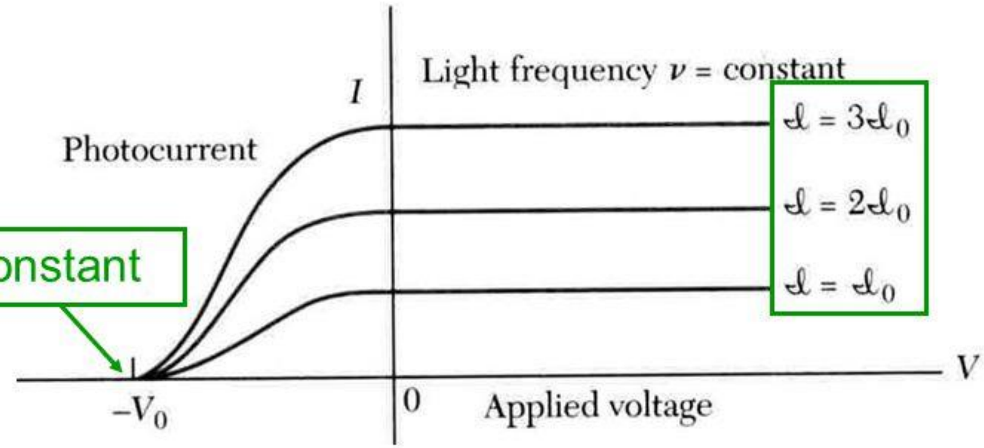
Photoelectric Effect: IV Curve Dependence

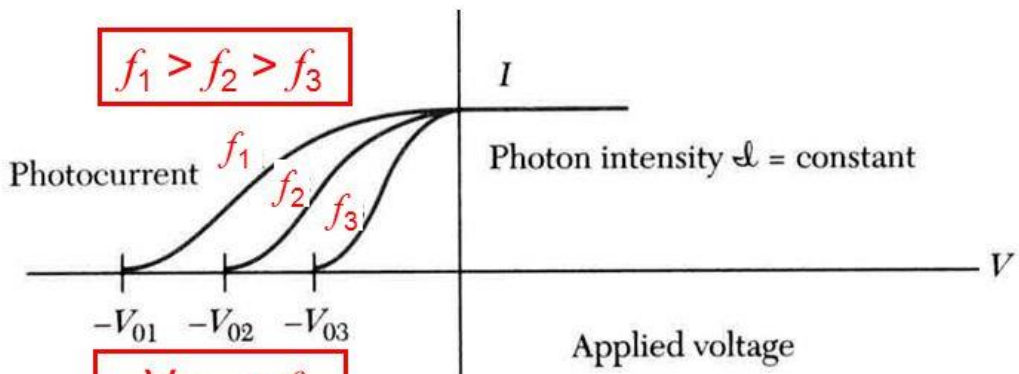
Intensity I dependence

V_{stop} = Constant



Frequency f dependence

$f_1 > f_2 > f_3$



$V_{stop} \propto f$

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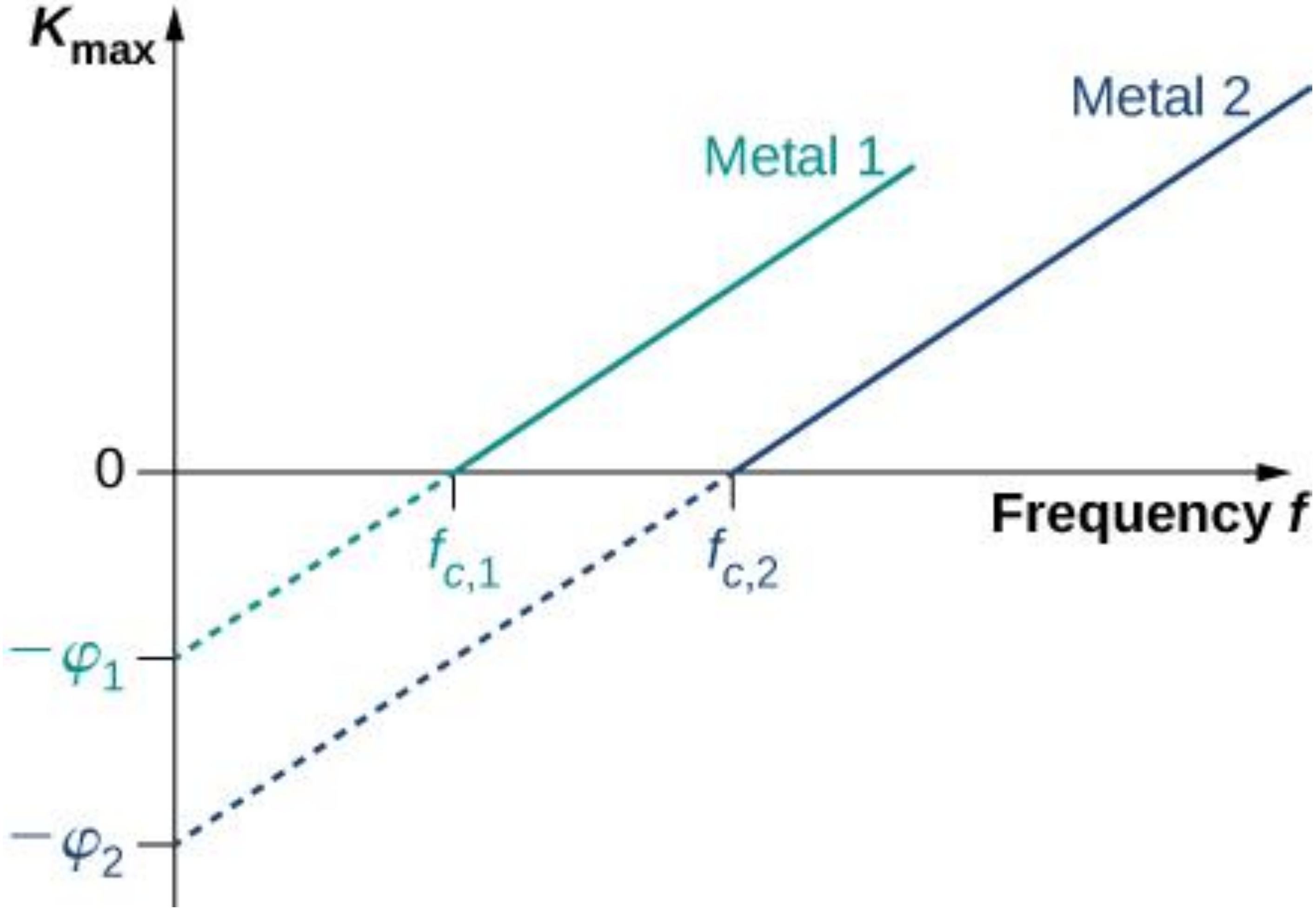

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Failure of Classical Wave Theory

According to classical wave theory,

- Intensity of a wave is the energy incident per unit area per unit time.
- Energy carried by an electromagnetic wave is proportional to the square of the amplitude of the wave.

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Explanation with classical theory

1. Existence of the threshold frequency

If intensity is sufficient , the electrons would absorb enough energy to escape. There should not be any threshold frequency.

2. Almost immediate emission of photoelectrons

Electrons require a period of time before sufficient energy is absorbed for it to escape from the metal. A bright light would eject electrons faster than a dim light.

3. The independence of kinetic energy of photoelectron on intensity and the dependence on frequency

According to classical wave theory, greater the intensity, the larger the energy of the light wave, so electrons are ejected with greater kinetic energy. It cannot explain why maximum kinetic energy is dependent on the frequency and independent of intensity.

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Einstein's explanation

- All electromagnetic radiation, is formed in discrete 'packet' of energy(or photon).
- Energy of a single photon is hf, h is Planck constant, f is frequency

When light is directed at a metal surface,

- Any free electron near the surface could be struck by a photon and gains energy
- If the gain in energy is sufficient to overcome the electromagnetic force of attraction between the electrons and the positive nucleus, the electron can leave the plate
- Each photo electron from a metal plate has gained the whole amount of energy of a single photon. The photon will be reflected or transmitted, if not absorbed. The photon's energy cannot be shared among the electrons

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Einstein's Photoelectric Equation

- Photon energy - **hf**
- Work function **- ϕ**
- Kinetic energy - **E**
- By law of energy conservation - **hf= E+ϕ**

$$E = hf - \phi$$

- Work function - **minimum amount of the work necessary to remove a free electron from the surface of the material.**

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Explanation…

- **The energy of the electrons does NOT depend on the intensity of the light.**

  Each electron absorbs only one photon at a time. If the absorbed energy is large enough to expel the electron from the metal, it leaves. If not, the electron dissipates its energy in collisions with nearby electrons and atoms before it can absorb another photon. And yes, this implies that the time it takes for an electron to lose the energy gained in one absorption is much smaller than the interval between absorptions. Under ordinary circumstances, it is.

- **The electrons always appear AS SOON AS the light reaches the plate (though a feeble light produces only a few).**

  As soon as a single photon containing sufficient energy strikes the source plate, it will knock an electron free. There is no need to wait for multiple waves to build up enough energy.

- **NO electrons are produced if the frequency of the light waves is below a critical value.**

  Electrons can only escape if the maximum kinetic energy is greater than zero.

  Ekmax >0   hf- ϕ > 0   f0 > ϕ /h

  For photoelectric effect to take place, *f >f0*

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- Simulation of Photoelectric Effect.

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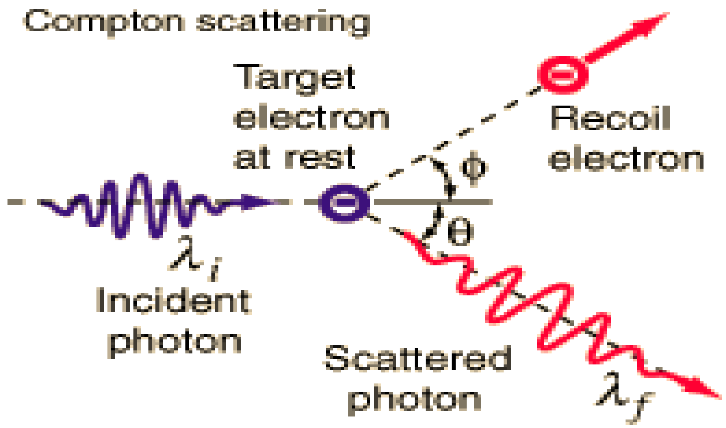
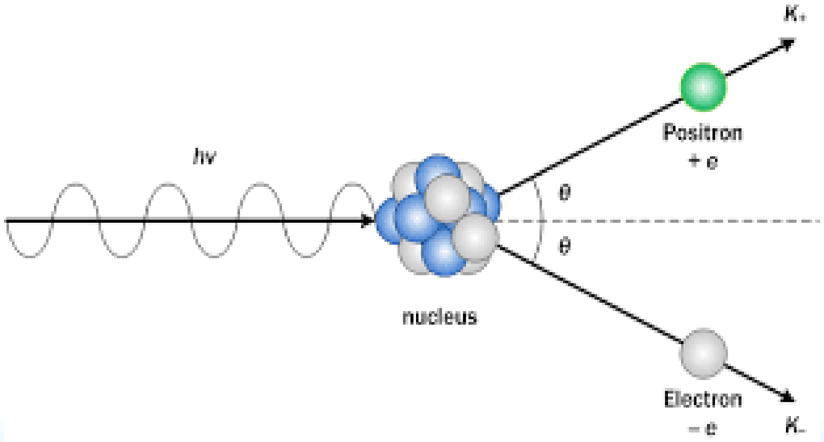
Interaction of matter and radiation

- Photoelectric Effect
- Compton Effect



- Pair Production



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